DEVOPS LABORATORY · BE COMPUTER ENGINEERING · AY 2026–27

Containerized Blood Bank Donor Matching Portal

A clinical donor–recipient matching service carried through a complete DevOps lifecycle: from problem definition and agile planning to a Jenkins pipeline that takes a commit to a running, health-verified container — and an Ansible layer that can roll it back.

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THE DELIVERY PIPELINE — NINE STAGES, PIPELINE-AS-CODE

01	02	03	04	05	06	07 GATE	08	09
Checkout	Build	Unit Test	Package	Deploy Tomcat	Health Check	UI Test Selenium	Promote to Registry	Post-deploy Verify

Declarative `Jenkinsfile` · parameterised port, image, profile · SCM-polled

101 s commit → verified container

ARCHITECTURE

Git (local)
48 commits · 19 tags · feature branches

→ **Jenkins LTS**
freestyle job + declarative pipeline

Maven 3.10 → `bloodbank.war`
Java 21 · Spring Boot 3.3.5 · WAR packaging

Multi-stage Docker image
build layer discarded → Tomcat 10.1 runtime, non-root uid 999

Private registry `:5001`
immutable build-number tags — the rollback source

→ **Ansible**
3 roles · idempotent · rollback

Running container `:9090`
H2 file DB on a named volume · `/actuator/health`

TOOLCHAIN & APPLICATION

LANGUAGE	**Java 21** (Temurin), Maven 3.10
FRAMEWORK	**Spring Boot 3.3.5** — MVC, Thymeleaf, Data JPA, Security
DATABASE	**H2 2.x** in file mode, persisted on a Docker named volume
RUNTIME	**Tomcat 10.1** — the WAR is a real servlet deployment
CI / CD	**Jenkins LTS 2.568.2** — freestyle job + `Jenkinsfile`
TESTING	**JUnit 5** (Surefire) + **Selenium 4** (Failsafe, headless Chrome)
CONTAINERS	**Docker 28.1.1**, multi-stage build, `registry:2`
CONFIG MGMT	**Ansible 2.21.5** — `community.docker`, 3 roles
AUTOMATION	**12 shell scripts** — one-command end-to-end demo

Domain logic. ABO/Rh compatibility is derived from antigens rather than hard-coded, donor eligibility is computed from a 90-day interval against an injected `Clock`, and the request workflow `NEW → MATCHED → FULFILLED | REJECTED` is enforced with role checks before transition legality.

VERIFIED RESULTS — EVERY FIGURE FROM A REAL RUN

195	64/64	101s	0	12
AUTOMATED TESTS 179 UNIT + 16 UI	COMPATIBILITY PAIRS PROVEN	COMMIT TO LIVE CONTAINER	ANSIBLE CHANGES ON RE-RUN	DEFECTS FOUND AND FIXED

FIFTEEN LIFECYCLE TASKS — ONE TAGGED COMMIT EACH

1. **Problem definition** & MVP scope freeze
2. **Agile backlog**, Kanban, Definition of Done
3. **SRS**, architecture & data model
4. **Git workflow**, repo & branch policy
5. **Feature branch** → reviewed merge
6. **MVP complete** + merge-conflict drill
7. **Jenkins CI** job, artefact archived
8. **Pipeline-as-code** & Tomcat deploy
9. **Selenium** journeys & test plan
10. **Quality gate** — a red build blocks deploy
11. **Dockerfile** & container lifecycle
12. **CD to registry**, versioned image
13. **Ansible** inventory & playbooks
14. **Idempotency**, health check, rollback
15. **Release v1.3.0**, report & viva guide

WHAT THE PROJECT ACTUALLY PROVES

- ✓ **The quality gate is real.** A deliberately removed authorisation check passed all 179 unit tests; Selenium caught it, the build went red, and Promote was never entered.
- ✓ **Rollback works under a minute.** Image tag 17 → 15 pulled from the private registry, container replaced, data intact on the volume, verified from the Docker API rather than from the app's own claim.
- ✓ **Health is behavioural.** Nine assertions — create a request, match it, fulfil it — not just an HTTP 200 from `/actuator/health`.
- ✓ **Idempotency is measured, not claimed.** A second `ansible-playbook site.yml` reports `changed=0` — and the check names the offending task if it ever does not.
- ✓ **The image is hardened.** The build stage is discarded, every stock Tomcat webapp removed, and the JVM runs as uid 999 as PID 1 after the entrypoint drops privileges.
- ✓ **One command reproduces everything.** `scripts/demo-all.sh` runs nine asserting steps commit-to-container in 151 s.